

Code-Layout, Readability and Reusability

Date	0
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S. No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform 4-space indentation used throughout the project	Yes	Follows Python PEP 8 standards
2	Proper File Structure	Code organized into modules such as app, routers, services, config, frontend	Yes	Easy to navigate and maintain
3	Meaningful Variable Names	Variables and functions use descriptive names	Yes	Improves readability
4	Function / Method Names	Functions are clearly named according to their purpose	Yes	Easy to understand
5	Code Comments	Important functions contain comments and docstrings	Yes	Key modules documented
6	Modular Design	AI, API, frontend and configuration are separated into modules	Yes	High modularity
7	No Redundant Code	Duplicate logic minimized through reusable functions	Yes	Better maintainability
8	Error Handling	Input validation and exception handling implemented	Yes	Prevents runtime failures

Reusable Components / Modules:

S. No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Event Analysis Module	Python + Transformers	Theme extraction	High

2	Conversation Generator	Python + GPT-2	Conversation generation	High
3	Fact Verification Module	Python + Wikipedia API	Fact validation	High
4	FastAPI Endpoints	FastAPI	Backend services	Medium
5	Configuration Module (config.py)	Python	Stores project configuration	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized modular project structure
Readability	5	Clear naming conventions and consistent formatting
Reusability	5	Independent modules reused across the application
Documentation / Comments	4	Important modules documented with comments and docstrings
Overall Score	4.8	Maintainable, modular, and easy to extend